

Alex Santora

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EDUCATION

Stevens Institute of Technology

Hoboken, NJ

B.E. Computer Engineering; B.S. Applied Mathematics

Sep. 2025 – May 2029

- **GPA: 3.88/4.00**; Dean's List; Edwin A. Stevens Scholarship; NCAA varsity baseball student-athlete.
- **Coursework:** Data Structures and Algorithms, Microprocessor Systems, Multi-variable Calculus, Linear Algebra

EXPERIENCE

University of New Hampshire

Durham, NH

Undergraduate Research Intern

May 2026 – Present

- Researched audio-based and physiological pain detection methods across 10+ peer-reviewed studies, identifying candidate ML features such as pitch, jitter, shimmer, heart-rate variability, and privacy-sensitive speech patterns.
- Developed a privacy-aware pain recognition workflow for extracting features, training candidate ML models, evaluating performance, and validating results under latency and privacy constraints.
- Translated literature findings into a prototype design for audio based pain detection, defining model inputs, evaluation metrics, and experimental requirements for future implementation.

Playland Arcade

Hampton Beach, NH

Arcade Attendant, Seasonal

May 2023 – Present

- Protected daily arcade operations by reconciling up to \$5,000 in cash transactions, managing prize redemption, resolving payment exceptions, and minimizing downtime across 100+ machines through basic troubleshooting and repair escalation.

PROJECTS

Baseball Pitch Classification Model | *Python, Pandas, Scikit-Learn, Joblib, Pybaseball*

Jun. 2026

- Achieved 93.98% pitch-type classification accuracy and 0.94 weighted F1 on an 87,113-pitch test set by training a balanced Random Forest on 350K+ Statcast-style pitch records.
- Improved class separation for overlapping pitch profiles by engineering features from velocity, spin rate, spin axis, vertical/horizontal break, release point, and extension.
- Built a reproducible ML workflow by automating data ingestion, preprocessing, model training, persistence, and evaluation using pandas, scikit-learn, joblib, and pybaseball.

Poker Neural Network | *Python, PyTorch, Reinforcement Learning, Neural Networks*

Jun. 2026

- Browser-based heads-up Texas Hold'em game with a neural network AI trained through reinforcement learning on 2M+ simulated hands.
- Improved AI decision quality by encoding 52-card deck state, hole cards, board cards, betting context, and hand-strength signals into neural-network-ready model inputs.

Autonomous Robot | *Arduino/C++, ESP8266, MQTT, Ultrasonic Sensors, SolidWorks*

Jan. 2026

- Reached 4 sequential navigation targets in a engineering project by integrating a WeMos ESP8266, 3 ultrasonic sensors, 2 DC motors, an OLED display, and embedded control logic.
- Improved obstacle-avoidance reliability by consuming MQTT x-y coordinate feedback, applying real-time decision logic, and tuning motor commands during road tests.

LEADERSHIP

NCAA Varsity Baseball

Stevens Institute of Technology

Student-Athlete; Leadership Council Member

Sep. 2025 – Present

- Balanced NCAA varsity competition with a rigorous engineering and applied mathematics course load by applying disciplined time management, consistent preparation, and team-first communication.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, Java, C#, C++, Arduino/C++

Machine Learning/Data: pandas, NumPy, scikit-learn, joblib, matplotlib, pybaseball, reinforcement learning, feature engineering, model evaluation

Engineering Tools: Git, SolidWorks, MQTT, ESP8266, ultrasonic sensing, embedded systems, browser-based applications